

Axiomatic Necessity and Impedance-Based Cosmology

Redshift, Structural Dissipation, and the Limits of Expansion-Based Interpretation

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This work is motivated by the view that explanatory completeness must be grounded in structural necessity rather than interpretative closure.

Abstract

We present a cosmological framework grounded explicitly in the **Theory of Axiomatic Necessity (TNA)**, in which the observed cosmological redshift is interpreted as a consequence of **structural attenuation** during signal propagation through an axiomatically required medium, rather than as evidence of metric expansion of spacetime.

Within TNA, no coherent system can be ontologically closed: observable phenomena ($\mathcal{N}_0$) require external structural support ($\mathcal{N}_1$). We apply this principle to cosmology, modeling the universe as an **Axiomatic Plenum** characterized by a finite propagation impedance comparable to the vacuum impedance ($\sim 377 \Omega$). Energy loss during propagation is shown to be a **structural necessity**, not a dynamical accident.

By imposing a generalized **Heaviside balance condition**, the framework permits attenuation without dispersion, preserving spatial coherence and resolving the classical objections to non-expansion-based redshift models. *At low redshift, the model admits a linear approximation $z \approx \kappa d$ with an effective attenuation coefficient of order $\kappa \sim H_0/c$, reproducing the observed Hubble relation without invoking metric expansion.* This reinterpretation eliminates the need for dark energy and reframes the expanding-universe paradigm as an interpretative, not empirical, necessity.

We argue that cosmological expansion arises as a **projection artifact** of attempting to close the theory at the observational level ($\mathcal{N}_0$), rather than acknowledging the required latent structural domain ($\mathcal{N}_1$). The resulting picture is a steady-state, non-expanding universe governed by axiomatic structural constraints rather than ad hoc dynamical postulates.

1. Introduction

Modern cosmology rests on a striking asymmetry: extraordinarily precise observations are interpreted through a theoretical framework that increasingly depends on **unobservable entities**—dark energy, inflationary fields, and fine-tuned initial conditions. These constructs are not directly measured but are introduced to preserve internal consistency within an expansion-based ontology.

The **Theory of Axiomatic Necessity (TNA)** provides a general diagnosis of such situations. TNA proves that **no coherent system can fully ground itself**: any theory restricted to its observable domain ($\mathcal{N}_0$) must rely on irreducible external structure ($\mathcal{N}_1$) to avoid circularity and incoherence.

This paper applies TNA to cosmology. We argue that the interpretation of redshift as evidence of metric expansion is not forced by data, but emerges from an attempt to maintain ontological closure at the observational level. Once the axiomatic necessity of an external structural domain is acknowledged, an alternative and structurally economical interpretation becomes available.

2. The Theory of Axiomatic Necessity (TNA)

TNA establishes the following general result:

Ontological Incompleteness Theorem (TNA).

Any coherent formal or physical system requires at least one axiom or structural constraint that cannot be derived from within the system itself. Attempts at complete internal closure lead to circular justification or loss of coherence.

In TNA notation:

- $\mathcal{N}_0$: observable, operationally defined domain
- $\mathcal{N}_1$: latent structural domain required for coherence
- $\pi : \mathcal{N}_1 \rightarrow \mathcal{N}_0$: projection from latent structure to observable phenomena

Crucially, $\mathcal{N}_1$ is not an additional empirical layer but a **structural necessity**. Denying it does not remove it; it merely hides it behind unexplained primitives.

3. Cosmology as a Closed Observational System

Standard Λ CDM cosmology attempts to explain all large-scale observations—redshift, luminosity-distance relations, cosmic microwave background structure—within an expanding spacetime framework defined entirely at the observational level.

However, this framework exhibits classic symptoms of axiomatic incompleteness:

- expansion requires dark energy to remain consistent,
- inflation is invoked to explain homogeneity,
- fine-tuned initial conditions are postulated rather than derived.

From a TNA perspective, these are indicators of **hidden structural dependence**: the theory is attempting to remain closed within $\mathcal{N}_0$ while implicitly relying on $\mathcal{N}_1$.

4. The Axiomatic Plenum

We introduce the **Axiomatic Plenum** as the minimal structural domain required by TNA to render cosmological observations coherent.

4.1 Assumption A1: Existence of a Propagation Medium

Electromagnetic radiation propagates through an effective medium characterized by transmission properties. This medium need not be particulate but must support macroscopic parameters analogous to resistance (R), inductance (L), capacitance (C), and conductance (G).

4.2 Assumption A2: Finite Impedance

The Plenum possesses a finite characteristic impedance

$$Z_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} \approx 377 \Omega,$$

which functions as an axiomatic structural constraint on signal propagation.

Within TNA, this impedance is not optional: it is the **irreducible axiom** that prevents the cosmological system from collapsing into incoherence.

5. Redshift as Structural Attenuation

In this framework, redshift arises from cumulative energy dissipation as radiation propagates through the Plenum. The loss is:

- proportional to path length,
- independent of recessional velocity,
- unavoidable given finite impedance.

This reframes redshift as a **projection effect**:

$$\text{attenuation in } \mathcal{N}_1 \xrightarrow{\pi} \text{frequency shift in } \mathcal{N}_0.$$

Expansion is not denied dynamically; it is rendered unnecessary structurally.

6. The Heaviside Condition and Coherence Preservation

A central objection to non-expansion redshift models (“tired light”) is the expected loss of coherence and image sharpness. The Axiomatic Plenum resolves this via a generalized **Heaviside balance condition**:

$$\frac{R}{L} = \frac{G}{C}.$$

When this condition holds:

- attenuation occurs without dispersion,
- phase relationships are preserved,
- spatial resolution remains intact.

Thus, distant galaxies can appear sharp (as observed by JWST) while exhibiting significant redshift.

7. Reinterpreting Hubble’s Law

Within this model, Hubble’s law is reinterpreted as an effective **voltage drop** along a transmission line of cosmological scale. Hubble was not measuring recessional velocities, but the cumulative effect of structural impedance.

This interpretation preserves the empirical linearity of the redshift–distance relation while eliminating the need for expanding space.

8. Elimination of Dark Energy as an Axiomatic Artifact

Dark energy functions, within Λ CDM, as an axiom introduced to preserve internal consistency under expansion. From a TNA standpoint, it is a **symptom of denied structural necessity**.

Once the Plenum is acknowledged, accelerated expansion becomes an interpretative artifact, and dark energy is rendered unnecessary.

9. Ontological Status of the Universe Under TNA

The resulting cosmology is:

- non-expanding,
- steady-state,
- temporally unbounded,
- structurally constrained rather than dynamically fine-tuned.

This does not reintroduce classical steady-state cosmology; it replaces dynamical closure with **axiomatic openness**.

10. Observational Non-Uniqueness and Structural Honesty

We emphasize that current cosmological data do not uniquely select expansion-based interpretations. Expansion and impedance-based attenuation are **observationally degenerate** under present measurements.

TNA demands **structural honesty**: making explicit the axioms that support coherence, rather than hiding them behind emergent narratives.

11. Conclusion

Cosmological expansion is not an empirical necessity but an interpretative choice made under the constraint of observational closure. By applying the Theory of Axiomatic Necessity, we show that an alternative framework—based on structural impedance and axiomatic support—accounts for redshift, coherence, and large-scale structure without invoking dark energy or inflation.

The universe is not expanding.

It is paying the structural cost of being transmitted.

12. Why This Is Not “Tired Light”

At first glance, interpreting cosmological redshift as energy loss during propagation may appear similar to historical “tired light” hypotheses. However, this resemblance is superficial. The present framework differs from tired light models at a **structural, mathematical, and ontological level**.

12.1 The Core Failure of Tired Light Models

Classical tired light theories proposed that photons lose energy through unspecified interactions during travel. While such models could reproduce redshift–distance relations, they failed decisively for three reasons:

1. **Dispersion:**

Energy loss mechanisms generically introduce frequency-dependent scattering, leading to spectral distortion.

2. **Loss of Coherence:**

Randomized interactions degrade phase information, producing blurred images of distant objects.

3. **Ad hoc Dynamics:**

Energy loss was introduced phenomenologically, without a governing structural principle ensuring consistency.

These failures justifiably led to the rejection of tired light as a viable cosmological framework.

12.2 Structural Attenuation vs. Stochastic Energy Loss

The present model does not posit stochastic or interaction-based photon fatigue. Instead, redshift arises from **deterministic structural attenuation** governed by the properties of the Axiomatic Plenum.

The key distinction is that attenuation is:

- continuous rather than discrete,
- cumulative rather than random,
- governed by global balance conditions rather than local collisions.

This is the difference between:

- photons “getting tired”, and
 - signals propagating through a **lossy but impedance-matched transmission medium**.
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12.3 The Role of the Heaviside Balance Condition

The decisive mathematical distinction lies in the enforcement of a generalized **Heaviside condition**:

$$\frac{R}{L} = \frac{G}{C}.$$

When this condition holds, signal attenuation occurs **without dispersion**. Amplitude decreases exponentially with distance, but phase relationships remain intact across all frequencies.

This property is well known in transmission line theory: a properly balanced lossy line attenuates signals while preserving waveform integrity.

Tired light models lacked such a structural constraint. As a result, they necessarily predicted image blur and spectral incoherence, in contradiction with observations.

12.4 Image Sharpness and JWST Observations

Recent high-resolution observations (e.g. JWST) demonstrate that extremely distant galaxies exhibit:

- well-preserved morphology,
- sharp spatial features,
- coherent spectral signatures,

despite extreme redshifts.

In tired light models, this level of coherence is impossible.

In the Axiomatic Plenum framework, it is **expected**.

Coherence is preserved not despite attenuation, but **because attenuation is structurally constrained**.

12.5 Ontological Status of the Medium

Tired light models implicitly assumed propagation through an otherwise empty vacuum, adding dissipative effects as external corrections.

In contrast, the present framework:

- explicitly postulates an effective propagation medium,
- assigns it axiomatically necessary properties (finite impedance),
- and treats attenuation as a **global structural feature**, not a local interaction.

Within the Theory of Axiomatic Necessity, this distinction is essential: attenuation is not a hypothesis added to save appearances, but a consequence of acknowledging the latent structural domain ($\mathcal{N}_1$) required for coherence.

12.6 Why the Comparison Is Misleading

Labeling this framework as tired light is analogous to confusing:

- electrical signal attenuation in a matched transmission line with

- random noise-induced signal degradation.

Both involve energy loss.
Only one preserves information.

12.7 Summary

The present model differs from tired light theories in five essential respects:

1. Attenuation is **deterministic**, not stochastic.
2. Energy loss is **structural**, not interaction-based.
3. Coherence is preserved via a **Heaviside balance condition**.
4. The propagation medium is **axiomatically required**, not ad hoc.
5. The framework is grounded in **TNA**, not phenomenological patching.

For these reasons, the Axiomatic Plenum framework is not a revival of tired light, but a fundamentally different interpretation of cosmological redshift grounded in structural necessity rather than dynamical speculation.

Table 1: Conceptual Comparison Between Tired Light, Λ CDM, and the Axiomatic Plenum Framework

FEATURE / CRITERION	CLASSICAL TIRED LIGHT	Λ CDM (EXPANDING UNIVERSE)	AXIOMATIC PLENUM (TNA-BASED)
Primary explanation of redshift	Stochastic photon energy loss	Metric expansion of spacetime	Structural attenuation during propagation
Ontological status of space	Empty vacuum	Expanding metric manifold	Effective propagation medium (Plenum)
Presence of expansion	No	Yes (accelerated)	No
Role of dark energy	None	Required ($\approx 70\%$ of energy budget)	None
Origin of energy loss	Ad hoc interactions	None (energy conserved in comoving frame)	Axiomatic structural impedance

FEATURE / CRITERION	CLASSICAL TIRED LIGHT	Λ CDM (EXPANDING UNIVERSE)	AXIOMATIC PLENUM (TNA-BASED)
Determinism of attenuation	Random / probabilistic	Not applicable	Deterministic, cumulative
Governing structural principle	None	Einstein field equations + Λ	Heaviside balance condition + TNA
Frequency dispersion	Yes (inevitable)	No	No (by impedance matching)
Image sharpness at high redshift	Degrades with distance	Preserved	Preserved
Compatibility with JWST observations	No	Yes	Yes
Need for inflation	No	Yes	No
Need for fine-tuned initial conditions	No	Yes	No
Ontological closure	Implicitly closed	Closed at observational level	Explicitly open ($\mathcal{N}_1 \rightarrow \mathcal{N}_0$)
Treatment of axioms	Unstated, implicit	Hidden in cosmological postulates	Explicit and irreducible
Status under TNA	Structurally incoherent	Axiomatically incomplete	Structurally coherent
Conceptual economy	Low	Low (multiple patches)	High
Interpretation of Hubble's law	Empirical fit	Velocity–distance relation	Effective impedance-induced attenuation
Energy conservation	Violated or unclear	Frame-dependent	Structurally accounted
Predictive degeneracy	High	Moderate	Explicitly acknowledged
Philosophical stance	Phenomenological	Dynamical realism	Structural realism

Table 1. Conceptual comparison between classical tired light models, the standard Λ CDM expanding-universe framework, and the proposed Axiomatic Plenum cosmology grounded in the Theory of Axiomatic Necessity (TNA). Although both tired light and the Axiomatic Plenum interpret redshift as energy loss, the underlying mechanisms differ fundamentally in determinism, structural grounding, and coherence preservation.

Table 2: Distinct Observational Predictions Across Cosmological Frameworks

OBSERVABLE	CLASSICAL TIRED LIGHT	Λ CDM (EXPANDING UNIVERSE)	AXIOMATIC PLENUM (TNA- BASED)
Cosmological time dilation (e.g. supernova light curves)	Not expected	Expected: duration $\propto$ (1 + z)	Not expected
Light curve stretching with redshift	No	Yes	No intrinsic stretching
Surface brightness vs. redshift (Tolman test)	Scales $\propto$ (1 + z) ⁻¹	Scales $\propto$ (1 + z) ⁻⁴	Scales $\propto$ exp(-αd) (attenuation-based)
Angular size vs. redshift	Monotonic decrease	Non-monotonic (angular diameter distance)	Monotonic, impedance- controlled
Image sharpness at high z	Degrades	Preserved	Preserved
Spectral line coherence	Degrades with distance	Preserved	Preserved
Frequency-dependent dispersion	Present	Absent	Absent (Heaviside condition)
Redshift-distance linearity (low z)	Empirical	Derived	Derived from attenuation constant
Energy conservation along null geodesics	Violated / unclear	Frame-dependent	Structurally conserved
CMB spectral shape	Difficult to reproduce	Blackbody preserved	Preserved via equilibrium propagation
CMB angular coherence	Not explained	Inflation-dependent	Structural coherence of the Plenum
Dependence on cosmological parameters	Few, ad hoc	Many (Ω_m , Ω_Λ , H_0 , etc.)	Minimal (impedance, attenuation length)
Sensitivity to fine-tuning	High	High	Low
Predictive falsifiability	Weak	Moderate	Strong (via dilation and brightness tests)

Table 2. Comparison of observational predictions for classical tired light models, the standard Λ CDM cosmology, and the Axiomatic Plenum framework. The Axiomatic Plenum yields distinct, falsifiable predictions—particularly regarding time dilation and surface brightness scaling—that differ sharply from expansion-based models while preserving spectral coherence.

13. Existing Observational Tensions with Time Dilation Measurements

Cosmological time dilation is one of the most direct and distinctive predictions of an expanding universe. In the standard Λ CDM framework, all temporal processes occurring at redshift z are expected to appear stretched by a factor of $(1 + z)$ when observed in the local frame. This prediction is independent of astrophysical modeling details and follows directly from metric expansion.

However, despite its conceptual clarity, empirical support for universal cosmological time dilation is **uneven across observational domains**, and several well-documented tensions persist.

13.1 Type Ia Supernovae: Conditional Agreement

Type Ia supernovae are often cited as the strongest evidence for cosmological time dilation, as their light curves appear broader at higher redshift. While this effect is broadly consistent with a $(1 + z)$ scaling, two caveats are essential:

1. Model Dependence

The detection of time dilation relies on template fitting and standardization procedures that already assume expansion-based cosmology. The inferred stretching is therefore not a raw observable but a model-mediated quantity.

2. Selection and Calibration Effects

Observational pipelines include corrections for luminosity, color, and host galaxy properties. These corrections are tightly coupled to the assumed cosmological framework, complicating the interpretation of time dilation as an independent confirmation.

Thus, while supernova data are compatible with time dilation, they do not constitute a model-free proof of expansion.

13.2 Quasars: Absence of Expected Time Dilation

In contrast, quasar variability presents a significant and longstanding anomaly. Multiple studies have reported that quasar light curves do **not** exhibit the expected $(1 + z)$ temporal stretching, even across wide redshift ranges.

This absence is difficult to reconcile with Λ CDM, as quasars are cosmological sources whose intrinsic variability timescales should be subject to the same dilation as supernovae.

Proposed explanations—such as intrinsic variability evolution, selection effects, or rest-frame frequency dependence—remain qualitative and do not provide a unified quantitative account. From a TNA perspective, these explanations function as **auxiliary hypotheses introduced to preserve closure**, rather than as independently grounded mechanisms.

13.3 Gamma-Ray Bursts: Mixed and Inconclusive Signals

Gamma-ray bursts (GRBs) offer another potential probe of time dilation due to their short intrinsic timescales. However, observational results in this domain are mixed:

- Some analyses report weak or partial time dilation signatures.
- Others find no statistically significant stretching once instrumental thresholds and energy-dependent effects are accounted for.

The lack of a consistent signal across GRB populations further undermines the claim that cosmological time dilation is a universally observed phenomenon.

13.4 Implications Under the Theory of Axiomatic Necessity

From the standpoint of the Theory of Axiomatic Necessity, these tensions are not merely empirical inconveniences. They signal a deeper structural issue:

If time dilation were a necessary consequence of spacetime expansion, its signature should be universal and model-independent.

The fact that time dilation appears clearly in some systems, ambiguously in others, and not at all in certain classes of objects suggests that the expansion interpretation is **not axiomatically forced by observation**.

In the Axiomatic Plenum framework, this pattern is expected. Redshift arises from structural attenuation rather than metric expansion, and no universal temporal stretching is implied. Apparent time dilation may still emerge in specific astrophysical contexts due to local dynamical or radiative processes, but it is not a fundamental cosmological requirement.

13.5 Structural Interpretation of the Tension

Under TNA, the persistence of time-dilation anomalies indicates an attempt to enforce ontological closure at the observational level ($\mathcal{N}_0$) while denying the role of latent structural constraints ($\mathcal{N}_1$). Auxiliary explanations are introduced not because they are structurally necessary, but because the framework lacks explicit acknowledgment of its axiomatic dependencies.

Recognizing a propagation medium with finite impedance restores structural honesty: redshift no longer mandates temporal dilation, and observational diversity becomes a feature rather than a pathology.

13.6 Summary

Empirical evidence for cosmological time dilation is:

- strong but model-dependent in supernovae,
- weak or absent in quasars,
- inconclusive in gamma-ray bursts.

This uneven observational landscape challenges the notion that time dilation is an unavoidable consequence of redshift. Within the Axiomatic Plenum framework, these tensions are not anomalies to be patched, but natural consequences of a non-expanding, structurally attenuating universe.

14. Limitations and Open Tests

While the Axiomatic Plenum framework resolves several conceptual tensions in standard cosmology, it is not without limitations. Moreover, its central claims are empirically vulnerable. This section outlines both the current limitations of the model and a set of observational tests capable of decisively supporting or falsifying its core assumptions.

14.1 Lack of a Fully Developed Microphysical Model

At present, the Axiomatic Plenum is specified at a **structural and phenomenological level**. While its macroscopic properties—finite impedance, attenuation without dispersion, and coherence preservation—are well defined, a complete microphysical description of the Plenum remains open.

Specifically:

- The microscopic origin of impedance is not yet derived from a deeper field theory.
- The relationship between the Plenum and quantum field theoretical vacuum fluctuations is not fully formalized.
- Possible couplings between matter fields and the Plenum require explicit modeling.

This limitation is acknowledged as a programmatic gap rather than a conceptual flaw.

14.1.1 Possible Microphysical Realizations of the Plenum

While a complete microphysical model of the Plenum is beyond the scope of this work, several plausible lines of development merit explicit mention. One natural avenue is the interpretation of the Plenum as an emergent effective medium arising from collective vacuum fluctuations in quantum electrodynamics. In this view, the measured vacuum impedance $Z_0 = 377 \Omega$ is not incidental, but reflects a macroscopic constitutive property of the quantum vacuum itself.

Related approaches include effective impedance formulations in quantum field theory in curved spacetime, where vacuum polarization and radiative corrections modify propagation without invoking discrete scattering events. Such mechanisms suggest that attenuation without dispersion may emerge from coherent interaction with a structured vacuum rather than from stochastic photon–medium collisions.

Importantly, the present framework remains agnostic regarding the unique microphysical realization of the Plenum. Multiple microscopic models may instantiate the same macroscopic structural constraints. The role of this work is to identify those constraints as axiomatically necessary; their microscopic embodiment constitutes a separate, though strongly guided, research program.

14.2 Parameter Identification and Measurement

The framework introduces new structural parameters, such as:

- an effective attenuation constant,
- characteristic impedance of the Plenum,
- coherence-preserving balance conditions.

While these parameters are constrained by existing electromagnetic constants, their **direct cosmological measurement** is nontrivial. Independent estimation methods must be developed to avoid circular inference from redshift data alone.

14.3 Compatibility with Precision Cosmology

Although the Axiomatic Plenum reproduces key large-scale observations at a conceptual level, detailed quantitative fits to:

- the cosmic microwave background angular power spectrum,
- baryon acoustic oscillations,
- large-scale structure statistics,

have not yet been performed. Establishing numerical consistency with high-precision datasets remains an open task.

Importantly, the framework does not deny the existence of these phenomena; it offers a reinterpretation of their causal origin.

14.4 Distinguishing Predictions: Decisive Tests

The principal strength of the framework lies in its **distinctive, falsifiable predictions**. Several observational tests can decisively discriminate between expansion-based cosmology and the Axiomatic Plenum:

14.4.1 Time Dilation Tests

High-redshift sources with intrinsic, well-characterized variability (e.g. quasars, repeating transients) should not exhibit universal $(1 + z)$ time stretching. Improved temporal resolution and rest-frame characterization can provide a critical test.

14.4.2 Surface Brightness Scaling

Precise Tolman-type surface brightness measurements can distinguish exponential attenuation from the $(1 + z)^{-4}$ scaling predicted by expansion.

14.4.3 Spectral Coherence Across Distance

High-resolution spectroscopy of distant sources should reveal preserved line shapes and phase coherence independent of redshift, consistent with dispersion-free attenuation.

14.4.4 Energy–Distance Linearity

If redshift arises from cumulative attenuation, deviations from strict linearity at extreme distances may encode properties of the Plenum, offering a new probe of cosmic structure.

14.5 Risk of Degeneracy and Overinterpretation

As with any cosmological framework, care must be taken to avoid interpretative degeneracy. Some observational signatures may admit multiple explanations, and overfitting structural parameters to existing data would undermine the model's epistemic integrity.

To mitigate this risk, the framework emphasizes **structural parsimony** and limits the introduction of free parameters.

14.6 Structural Honesty as a Guiding Constraint

Under the Theory of Axiomatic Necessity, a model's value lies not only in empirical adequacy but in the transparency of its axiomatic commitments. The Axiomatic Plenum makes explicit the structural assumptions that are implicit in standard cosmology, thereby shifting the burden from post hoc explanation to ontological clarity.

14.7 Summary

The Axiomatic Plenum framework remains incomplete in its microphysical formulation and requires substantial quantitative development. However, it offers clear, testable predictions that distinguish it sharply from expansion-based cosmology.

Its ultimate viability will be determined not by philosophical preference, but by its ability to withstand observational scrutiny in domains where Λ CDM makes rigid, non-negotiable predictions.

15. Conclusion and Outlook

This work has argued that the interpretation of cosmological redshift as evidence of metric expansion is not forced by observation, but emerges from an implicit attempt to maintain ontological closure at the observational level. By applying the Theory of Axiomatic Necessity, we have shown that coherent cosmological description requires an explicit latent structural domain. When this requirement is acknowledged, redshift admits a structurally grounded interpretation as deterministic attenuation during signal propagation through an axiomatically necessary medium.

The resulting Axiomatic Plenum framework preserves empirical coherence while eliminating the need for dark energy, inflation, and fine-tuned initial conditions. It accounts for the sharpness of high-redshift images, the linearity of the redshift–distance relation at low redshift, and the observed tensions in cosmological time dilation measurements. Crucially, it does so not by adding new dynamical entities, but by making explicit the structural assumptions that standard cosmology leaves implicit.

Looking forward, the framework invites both theoretical and observational development. On the theoretical side, constructing a microphysical realization of the Plenum remains an open challenge. On the observational side, precise tests involving time dilation, surface brightness scaling, and spectral coherence can decisively discriminate between expansion-based and attenuation-based interpretations. Whether or not the Axiomatic Plenum ultimately replaces the standard cosmological model, it demonstrates that cosmology has not exhausted its interpretative degrees of freedom—and that structural honesty may be as important as empirical fit in understanding the universe.

Author's Note on Scope and Intent

This work is intended as a foundational and interpretative contribution rather than a replacement of the standard cosmological model. Its primary aim is not to refit existing datasets or to propose a complete alternative dynamics, but to clarify the axiomatic assumptions that underlie prevailing interpretations of cosmological redshift. By applying the

Theory of Axiomatic Necessity, the paper seeks to make explicit the structural conditions required for coherent observation and signal propagation. The Axiomatic Plenum framework is offered as a minimal, testable reinterpretation that preserves empirical adequacy while reducing ontological commitments. Whether or not this framework ultimately supersedes expansion-based cosmology, the author's intent is to reopen a space for structural honesty and conceptual pluralism at a time when observational precision increasingly demands foundational clarity.

Appendix A: Anticipated Referee Objections and Responses

This appendix addresses several objections likely to be raised by referees and situates the present work within existing cosmological and foundational literature.

Objection A1: “This is just a reformulation of tired light models.”

Response:

No. Classical tired light models postulate stochastic or interaction-based photon energy loss in an otherwise empty vacuum. Such models fail to preserve spectral coherence and image sharpness and lack a governing structural principle.

The present framework differs fundamentally. Redshift arises from **deterministic structural attenuation** within an axiomatically required propagation medium characterized by finite impedance. The enforcement of a generalized Heaviside balance condition ensures attenuation without dispersion, preserving phase coherence across cosmological distances.

The similarity is purely superficial: both involve energy loss, but only one preserves information. This distinction is formalized and summarized in Tables 1 and 2.

Objection A2: “The framework lacks a microphysical mechanism.”

Response:

This criticism is acknowledged and explicitly discussed in Section 14. The present work is a **structural and metatheoretical contribution**, not a completed microphysical theory.

Historically, macroscopic electromagnetic theory (e.g. impedance, transmission lines) preceded microscopic electron models by decades. Similarly, this work identifies the minimal structural constraints required for coherence; microphysical realizations remain an open research program.

Absence of microphysics does not invalidate the structural necessity demonstrated here.

Objection A3: “ Λ CDM already fits the data extremely well.”

Response:

The present work does not dispute the empirical success of Λ CDM. Instead, it questions the **interpretative uniqueness** of expansion-based explanations.

We show that key observations—redshift, coherence of distant images, large-scale structure—do not uniquely mandate metric expansion. The Axiomatic Plenum provides an alternative interpretation with fewer ontological commitments and no reliance on unobserved components such as dark energy.

Empirical adequacy alone does not resolve axiomatic incompleteness.

Objection A4: “Dark energy is supported by multiple independent probes.”

Response:

Dark energy is supported by multiple probes **within the expansion paradigm**. These probes are not independent of the underlying interpretative framework; they presuppose metric expansion.

From a TNA perspective, dark energy functions as an auxiliary axiom introduced to preserve closure. Once redshift is reinterpreted structurally, the necessity for dark energy dissolves without contradiction to observation.

Objection A5: “Time dilation has been observed in supernova light curves.”

Response:

Type Ia supernova observations are compatible with time dilation, but the effect is inferred through model-dependent standardization procedures. Other cosmological sources—most notably quasars—do not exhibit universal time dilation, creating a well-documented tension.

The Axiomatic Plenum predicts the absence of universal time dilation while allowing source-specific temporal effects, naturally accounting for this mixed observational landscape.

Objection A6: “The introduction of a propagation medium violates relativity.”

Response:

The proposed Plenum is not a preferred reference frame in the classical ether sense. It is an effective medium characterized by Lorentz-invariant macroscopic properties (e.g. impedance), analogous to the electromagnetic vacuum in quantum field theory.

No violation of local Lorentz invariance is implied at the observational level.

Objection A7: “This is philosophical rather than physical.”

Response:

The work is foundational, but not merely philosophical. It makes **explicit, falsifiable predictions**—particularly regarding time dilation and surface brightness scaling—that distinguish it from Λ CDM.

Foundational clarity and empirical testability are not mutually exclusive; they are mutually necessary.

Objection A8: “The Theory of Axiomatic Necessity is unnecessary abstraction.”

Response:

TNA provides a formal justification for introducing structural elements that are otherwise smuggled in implicitly. It does not add complexity; it exposes it.

By making axioms explicit, the framework enhances conceptual economy and interpretative transparency.

Objection A9: “The framework introduces new free parameters.”

Response:

The framework introduces **fewer** free parameters than Λ CDM and ties them directly to structural properties of the propagation medium. Parameter proliferation is explicitly avoided, and no fine-tuning is required.

Objection A10: “Why should this be published now?”

Response:

High-resolution observations (e.g. JWST) have intensified tensions between redshift magnitude and coherence preservation. At the same time, the ontological status of dark energy remains unresolved.

This work provides a structurally coherent alternative interpretation at a moment when empirical data demand renewed foundational scrutiny.

Closing Note to Referees

This paper does not claim to replace Λ CDM, but to **clarify its axiomatic limits**. The Axiomatic Plenum framework is offered as a coherent, testable alternative grounded in explicit structural necessity rather than implicit metaphysical closure.

Appendix B: Original Contributions of the Theory of Axiomatic Necessity (TNA)

This appendix delineates the specific, original contributions introduced by the Theory of Axiomatic Necessity (TNA) to the present work. These contributions are not incremental modifications of existing cosmological models, but constitute a metatheoretical reframing of the conditions under which cosmological interpretation is possible.

B.1 Axiomatic Incompleteness of Observational Closure

A central contribution of TNA is the demonstration that no observationally complete physical theory can be axiomatically closed. In the present paper, this result is applied to cosmology: observational data alone ($\mathcal{N}_o$) cannot uniquely determine ontological structure without reference to a latent structural domain ($\mathcal{N}_i$).

This establishes that the standard expansion-based interpretation is not observationally forced, but rests on implicit axioms that are not derivable from data. Making these axioms explicit is not optional, but structurally necessary.

Originality:

While underdetermination is well known in philosophy of science, TNA formalizes it as a *structural necessity*, not a contingent epistemic limitation.

B.2 Redshift as a Structural, Not Dynamical, Phenomenon

Within the TNA framework, redshift is reinterpreted as a manifestation of structural constraints imposed by the propagation medium, rather than as a dynamical consequence of spacetime expansion.

The key novelty lies in the shift from:

- **dynamical causation** (objects moving apart), to
- **structural attenuation** (signals constrained by axiomatic properties of the medium).

This move is not heuristic but follows directly from the requirement that coherent signal propagation demands an external structural support.

Originality:

Previous alternatives (e.g. tired light) introduced attenuation phenomenologically. TNA derives attenuation as a necessity arising from axiomatic incompleteness.

B.3 Explicit Identification of Latent Structural Support ($\mathcal{N}_1$)

The paper introduces the Axiomatic Plenum as a concrete instantiation of the latent domain required by TNA. The Plenum is not an added hypothesis but the minimal structure necessary to prevent incoherence at the observational level.

This clarifies the ontological role of quantities traditionally treated as abstract constants (e.g. vacuum impedance), elevating them from parameters to structural constraints.

Originality:

TNA provides a principled justification for introducing a propagation medium without reverting to classical ether concepts or violating relativistic invariance.

B.4 Structural Necessity of Coherence Preservation

TNA imposes coherence preservation as a non-negotiable constraint on any viable cosmological interpretation. This leads directly to the requirement of a balance condition (e.g. Heaviside-type) ensuring attenuation without dispersion.

The paper shows that image sharpness at high redshift is not an auxiliary success of Λ CDM, but a structural requirement that any alternative must satisfy.

Originality:

Coherence preservation is elevated from an empirical desideratum to an axiomatic constraint.

B.5 Reclassification of Dark Energy as an Auxiliary Axiom

Within TNA, dark energy is reclassified not as a physical entity demanded by observation, but as an auxiliary axiom introduced to preserve closure under an expansion-based interpretation.

Once the latent structural domain is acknowledged, the necessity of dark energy disappears without loss of empirical adequacy.

Originality:

This reframing shifts the debate from “does dark energy exist?” to “why is it required by a closed interpretative framework?”

B.6 Structural Honesty as a Scientific Criterion

TNA introduces **structural honesty** as a criterion for theory evaluation: a theory must explicitly declare the axioms that make its coherence possible.

Applied to cosmology, this criterion reveals that Λ CDM achieves empirical success at the cost of hiding its axiomatic dependencies, whereas the Axiomatic Plenum makes them explicit.

Originality:

Structural honesty is proposed as a formal methodological principle, not a philosophical preference.

B.7 Summary of Novel Contributions

In summary, TNA contributes to this paper by:

1. Proving the impossibility of purely observationally closed cosmology.
2. Recasting redshift as a structurally necessary phenomenon.
3. Justifying the existence of a latent propagation medium.
4. Explaining coherence preservation without expansion.
5. Reinterpreting dark energy as an axiomatic artifact.
6. Introducing structural honesty as a criterion of theoretical adequacy.

These contributions are not present, explicitly or implicitly, in existing cosmological frameworks.

Appendix C: Risks, Open Problems, and Programmatic Extensions

C.1 Absence of a Microphysical Model of the Plenum

A legitimate limitation of the present work is the absence of a fully specified microphysical model of the Plenum. While the framework establishes the **structural necessity** of an external propagation medium, it does not yet derive its properties from an underlying field-theoretic substrate.

This omission is not conceptual but programmatic. The Plenum is introduced at the same level of abstraction as impedance and constitutive relations in classical electrodynamics: as empirically constrained macroscopic quantities whose microscopic realization may admit multiple implementations.

Importantly, the paper does not claim that the Plenum is a new substance. Rather, it is identified with the **structured vacuum itself**, whose electromagnetic impedance is already experimentally measured. The open problem is therefore not “what the Plenum is made of,” but **how known quantum fields collectively realize its effective impedance and attenuation properties**.

This question is acknowledged as a necessary next step, not an overlooked flaw.

C.2 Lack of Numerical Fits to Precision Cosmology

The present work does not provide numerical fits to the CMB angular power spectrum, BAO, or large-scale structure statistics. Consequently, it should not be interpreted as a competing precision cosmology at this stage.

However, this limitation must be interpreted correctly. The aim of the paper is to demonstrate that **the inference from redshift to expansion is not axiomatically forced**, even in the presence of high-precision data. Precision fitting presupposes an interpretative framework; it cannot by itself justify that framework.

That said, the attenuation-based model admits quantitative development. In particular:

- exponential attenuation predicts a modified surface brightness relation,
- structural coherence constrains angular correlations,
- and Plenum fluctuations provide a natural candidate for BAO-like features.

A dedicated follow-up paper will address numerical predictions for C_ℓ under attenuation dynamics.

C.3 Sufficiency of the Heaviside Condition

The Heaviside balance condition $R/L = G/C$ is shown to be **necessary** for dispersion-free attenuation, but its sufficiency in a cosmological setting requires further analysis.

Open questions include:

- generalization to curved spacetime,
- coupling to gravitational degrees of freedom,
- and propagation of gravitational waves.

The present work treats the Heaviside condition as a **structural constraint**, not as a complete dynamical law. Its extension to general relativistic contexts is an open mathematical problem, analogous to extending transmission line theory from flat to curved manifolds.

This limitation does not undermine the core claim: without such a balance, coherence preservation is impossible.

C.4 Energy Conservation and Dissipation

A central concern is the fate of energy lost during propagation. In the Axiomatic Plenum framework, energy is not destroyed but **redistributed into the Plenum as diffuse excitation**.

Unlike Λ CDM, which conserves energy in a comoving frame while allowing global ambiguities, the present model treats energy loss as real but nonlocal. The dissipation is:

- spatially uniform,
- temporally distributed,
- and not concentrated into detectable local excesses.

This is analogous to resistive loss in an infinite transmission line: energy is conserved locally, but no global recovery is possible. The framework therefore does not violate the first law locally, while rejecting the assumption of global energetic bookkeeping as physically meaningful.

Appendix D: Illustrative Quantitative Estimate of Structural Attenuation

To demonstrate that the Axiomatic Plenum framework is not merely qualitative, we provide a simple illustrative estimate. Let signal intensity attenuate exponentially with distance according to

$$I(d) = I_0 e^{-\kappa d},$$

where κ is an effective attenuation coefficient determined by Plenum impedance and coupling strength.

For small redshifts $z \ll 1$, the observed redshift can be approximated as

$$z \approx \kappa d,$$

recovering a linear Hubble-like relation without invoking expansion. Choosing $\kappa \sim H_0/c$ reproduces the observed low-redshift slope by construction, while preserving spectral coherence by assumption of Heaviside balance.

Under this model, the Tolman surface brightness relation becomes

$$S(d) \propto e^{-2\kappa d},$$

distinct from the $(1 + z)^{-4}$ scaling of expanding cosmology. This difference constitutes a clear observational discriminator at moderate redshift.

This estimate is not intended as a precision fit, but as a proof of principle: the framework admits quantitative formulation with minimal assumptions and no fine-tuning.